Design and LTspice Characterization of a Current-Starved Ring Voltage-Controlled Oscillator

Harshaditya Punera
M.Tech, Electrical Engineering
Indian Institute of Technology Kanpur

Abstract

This report documents the design and pre-layout simulation of a current-starved ring voltage-controlled oscillator (VCO) using LTspice and a TSMC 180 nm transistor-model file. Two oscillator configurations are investigated: a three-stage ring and a nine-stage ring. The control voltage V_{con} is swept to characterize frequency tuning and average supply power. Transient waveforms are inspected, and peak/minimum output voltages are measured over a steady-state window. A small sinusoidal perturbation on the control voltage is also explored qualitatively. The three-stage configuration reaches approximately 3.14 GHz at $V_{\mathrm{con}} = 1.8\,\mathrm{V}$, while the nine-stage configuration reaches approximately 1.136 GHz at the same control voltage. The nine-stage configuration reaches a lower frequency at the lower end of the sweep, approximately 226.8 MHz at 0.7 V. These results demonstrate voltage-controlled oscillation in the simulated circuit. The current results are schematic-level simulations; layout, parasitic extraction, formal noise/jitter characterization, and silicon measurements have not yet been completed.

Contents

1 Introduction **3**

2 Objectives **3**

3 Circuit Architecture and Simulation Environment **3**
3.1 Oscillator configurations . . . 3
3.2 Model and tools . . . 4
3.3 Transient analysis . . . 5

4 Measurement Methodology **5**
4.1 Oscillation frequency . . . 5
4.2 Average supply current and power . . . 5
4.3 Peak and minimum output voltage . . . 5

5 Frequency Tuning Results **6**
5.1 Frequency Tuning Characteristics of Different Stage Configurations . . . 6
5.2 Three-stage configuration . . . 7
5.3 Nine-stage configuration . . . 8
5.4 Frequency versus control voltage . . . 10

6 Power Consumption **10**
6.1 Three-stage configuration . . . 10
6.2 Nine-stage configuration . . . 10

7 Startup Delay Analysis **12**
7.1 Startup Delay Results 12
7.2 Startup Waveforms 12

8 Control-Voltage Perturbation Experiment **12**

9 Discussion **13**

1 Introduction

A voltage-controlled oscillator is a circuit whose oscillation frequency varies with a control voltage. VCOs are important building blocks in phase-locked loops (PLLs), clock-generation circuits, frequency synthesizers, and communication systems.

A ring oscillator uses an odd number of inverting stages connected in a loop. Oscillation occurs because the total loop phase shift reaches 360° at a frequency for which the loop gain is sufficient. A current-starved inverter limits the current charging and discharging the effective capacitance at each stage. Changing the control voltage changes the available current and therefore the propagation delay and oscillation frequency.

For a ring oscillator with N stages, a commonly used first-order approximation is

$$f_{\text{osc}} \approx \frac{1}{2Nt_d}, \tag{1}$$

where t_d is the average propagation delay per stage. This is an approximate relation; the actual frequency depends on transistor models, bias conditions, loading, and parasitic capacitances.

The goal of the present work is to build and characterize a current-starved ring VCO in LTspice before proceeding to a possible physical-design flow.

2 Objectives

The work carried out so far has the following objectives:

1. Construct a current-starved ring oscillator using MOS transistor models.
2. Support both three-stage and nine-stage oscillator configurations.
3. Sweep V_{con} over the range 0.7–1.8 V.
4. Measure oscillation frequency from transient waveforms.
5. Measure average supply current and estimate average supply power.
6. Inspect steady-state output waveforms and quantify voltage excursions beyond the nominal supply rails.
7. Perform an initial qualitative experiment with a sinusoidal perturbation on the control voltage.
8. Identify further work needed for output buffering, FFT analysis, and eventual layout-level validation.

3 Circuit Architecture and Simulation Environment

3.1 Oscillator configurations

The circuit has two selectable configurations:

- **Three-stage mode:** a three-stage current-starved inverter ring intended for higher-frequency operation.
- **Nine-stage mode:** a nine-stage current-starved inverter ring intended for lower-frequency operation.

The schematic uses the selection controls labelled `S3` and `S9` to select the corresponding configuration. The control voltage is parameterized as {`VCON`} for sweeping. The oscillator output is currently measured directly from the final oscillator stage; no output buffer has yet been added.

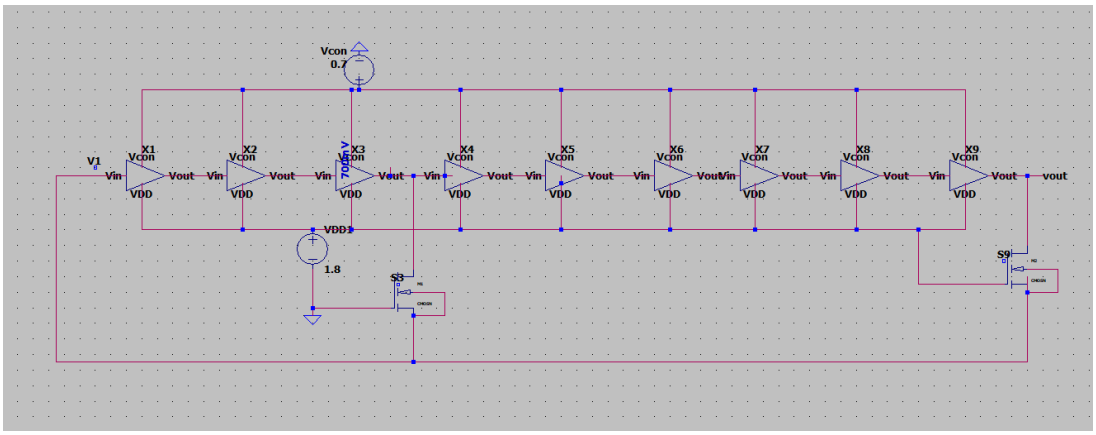


Figure 1: LTspice schematic of the current-starved ring VCO, showing the three-stage/nine-stage selection, control-voltage source, supply, and output node.

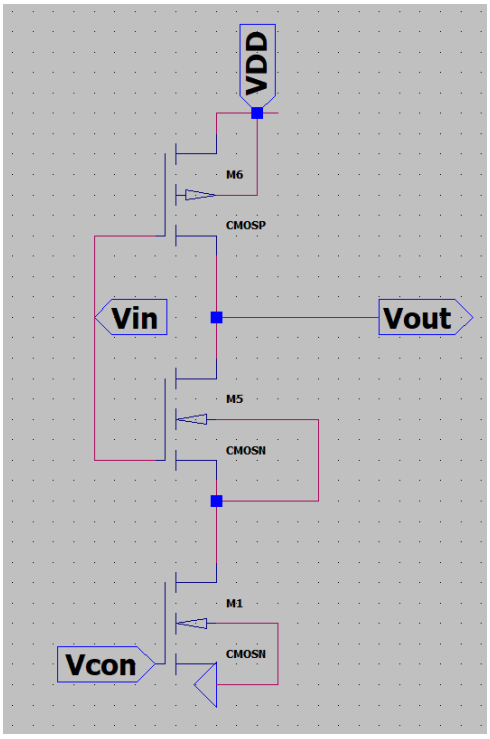


Figure 2: CMOS inverter/circuit diagram supplied with the project.

3.2 Model and tools

The simulations were performed in LTspice using the model file `tsmc018.lib`, referenced locally as `C:`
`textbackslash LT-Models`
`textbackslash tsmc018.lib`. The nominal supply used in the power calculation is $V_{\mathrm{DD}} = 1.8\,\mathrm{V}$. The output threshold used for period measurement is $0.9\,\mathrm{V}$, i.e. half of the nominal supply.

3.3 Transient analysis

A typical transient directive used for the sweep was

```
.tran 0 100n 0 2p
```

where the simulation stop time is 100 ns and the maximum timestep is 2 ps. Shorter transient runs were also used during debugging; measurement windows must always contain enough steady-state crossings for valid frequency measurements.

4 Measurement Methodology

4.1 Oscillation frequency

The period is measured between two rising crossings of the output at 0.9 V, separated by one cycle:

```
.meas tran Tperiod TRIG V(vout) VAL=0.9 RISE=10
+ TARG V(vout) VAL=0.9 RISE=11
.meas tran Freq PARAM 1/Tperiod
```

The frequency is calculated as

$$f_{\mathrm{osc}} = \frac{1}{T_{\mathrm{period}}}. \tag{2}$$

The measured frequency can depend slightly on the selected crossings and the transient timestep. Small differences between runs should therefore be interpreted in the context of the measurement setup.

4.2 Average supply current and power

The average current was measured from 20 ns to 100 ns, after allowing the oscillator to start:

```
.meas tran Iavg AVG -I(VDD1) FROM=20n TO=100n
.meas tran Pavg PARAM 1.8*Iavg
```

The supply source instance is named `VDD1`. The sign convention in the expression `-I(VDD1)` is chosen to report current drawn from the supply as a positive value. Average power is estimated as

$$P_{\mathrm{avg}} = V_{\mathrm{DD}} I_{\mathrm{avg}}, \tag{3}$$

using $V_{\mathrm{DD}} = 1.8\,\mathrm{V}$. This is a supply-power estimate based on the measured average current; it does not include any additional circuitry that may be added later, such as an output buffer.

4.3 Peak and minimum output voltage

The steady-state output extremes were measured over the interval from 50 ns to 100 ns:

```
.meas tran VMAX MAX V(vout) FROM=50n TO=100n
.meas tran VMIN MIN V(vout) FROM=50n TO=100n
```

The excursions relative to the nominal rails are calculated as

$$V_{\mathrm{over}} = \max(V_{\mathrm{max}} - V_{\mathrm{DD}}, 0), \tag{4}$$

$$V_{\mathrm{under}} = \max(0 - V_{\mathrm{min}}, 0). \tag{5}$$

These are measured extrema of an oscillator waveform, not a classical step-response overshoot measurement.

5 Frequency Tuning Results

5.1 Frequency Tuning Characteristics of Different Stage Configurations

Figure 3 compares the oscillation frequency as a function of control voltage for the 3-stage, 5-stage, 7-stage, and 9-stage ring oscillator configurations.

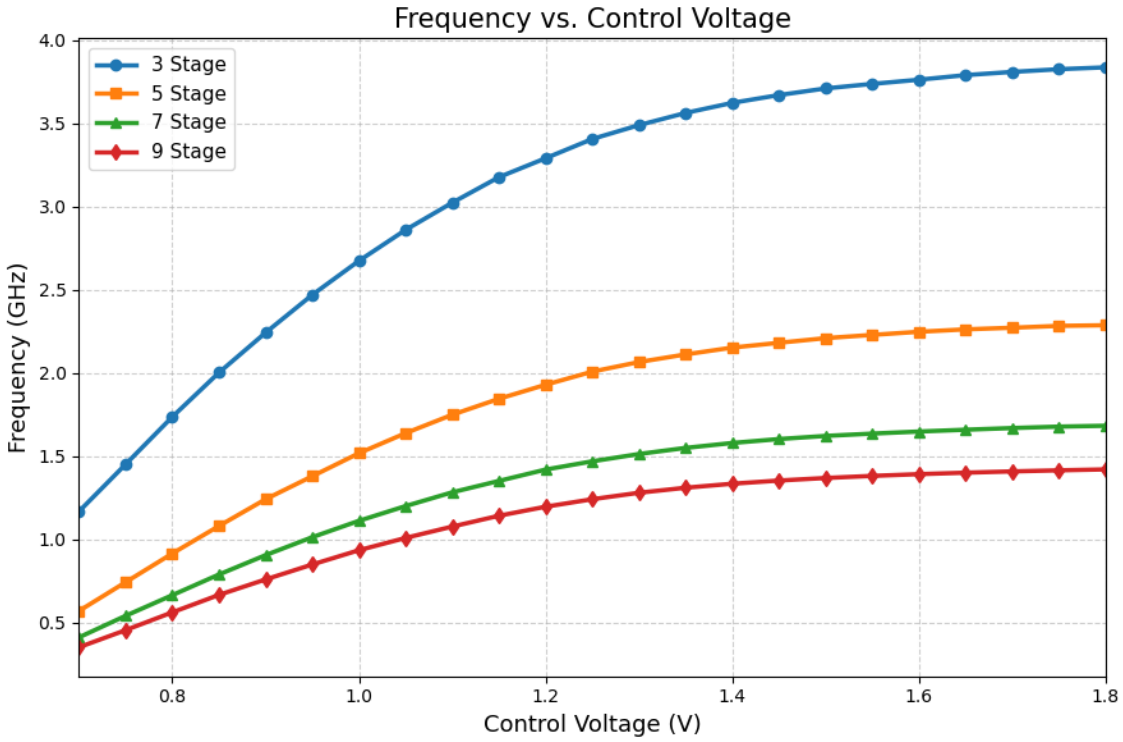


Figure 3: Frequency versus control voltage for different ring oscillator stage configurations.

As shown in Fig. 3, the oscillation frequency increases with the control voltage for all four configurations. Increasing the control voltage increases the available current in the current-starved inverter stages, thereby reducing the effective propagation delay and increasing the oscillation frequency.

The 3-stage configuration exhibits the highest oscillation frequency over the entire control-voltage range, reaching approximately 3.84 GHz at $V_{\text{con}} = 1.8\,\text{V}$. The 5-stage configuration reaches approximately 2.29 GHz, while the 7-stage and 9-stage configurations reach approximately 1.68 GHz and 1.42 GHz, respectively, at the same control voltage.

The observed trend is consistent with the approximate ring oscillator frequency relation

$$f_{\text{osc}} \approx \frac{1}{2Nt_d}, \tag{6}$$

where N is the number of inverter stages and t_d is the average propagation delay per stage. For a given control voltage, increasing the number of stages increases the total loop delay and consequently reduces the oscillation frequency.

The curves also exhibit a nonlinear relationship between frequency and control voltage. The frequency increment becomes smaller at higher control voltages, indicating a reduction in tuning sensitivity in this operating region. This behaviour is associated with the nonlinear current-voltage characteristics of MOS transistors and the dependence of propagation delay on the control voltage.

Overall, the comparison illustrates the effect of the number of inverter stages on the frequency range of the ring oscillator. The 3-stage configuration is suitable for higher-frequency

operation, whereas configurations with more stages provide lower-frequency operation over the same control-voltage range. ‘‘‘

5.2 Three-stage configuration

The three-stage oscillator was swept over $V_{\mathrm{con}} = 0.7$–$1.8\,\mathrm{V}$ in $0.01\,\mathrm{V}$ increments, giving 111 operating points. The measured frequency increased monotonically over the sweep. Representative results are shown in Table 1.

Table 1: Representative frequency results for the three-stage configuration.

V_{con} (V)	Frequency (MHz)	Frequency (GHz)
0.70	808.46	0.8085
0.80	1194.08	1.1941
0.90	1553.32	1.5533
1.00	1864.33	1.8643
1.10	2132.66	2.1327
1.20	2366.73	2.3667
1.30	2569.31	2.5693
1.40	2741.59	2.7416
1.50	2884.12	2.8841
1.60	2997.62	2.9976
1.70	3084.24	3.0842
1.80	3144.74	3.1447

The frequency rises from approximately 0.808 GHz to 3.145 GHz across the measured control-voltage range. The tuning response is nonlinear: the increase in frequency per 0.1 V becomes smaller toward the high-control-voltage end of the sweep.

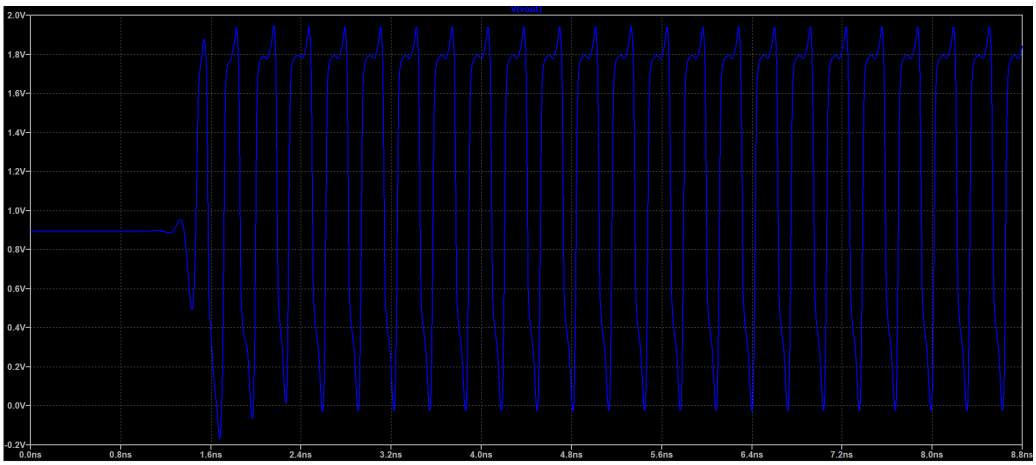


Figure 4: Transient output waveform of the three-stage VCO at 1.8V control voltage.

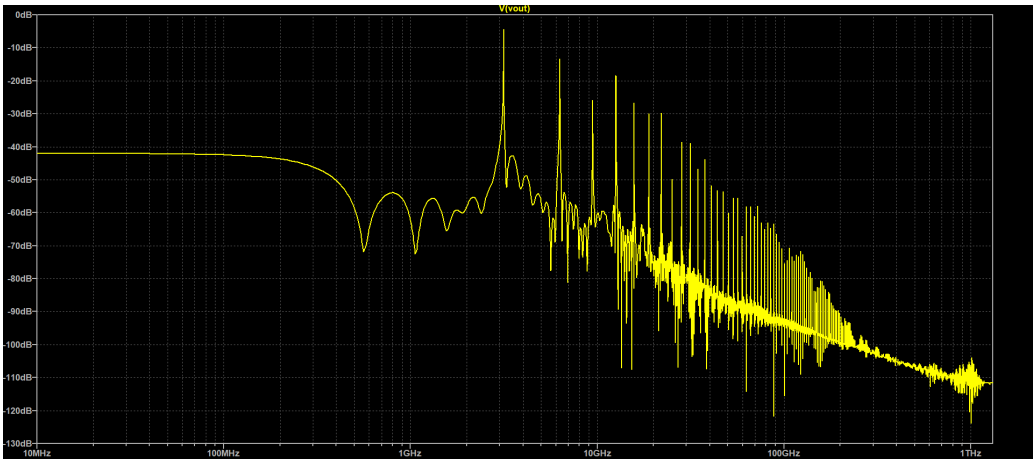


Figure 5: FFT analysis of the three-stage VCO output.

5.3 Nine-stage configuration

The nine-stage oscillator was characterized over the same overall control-voltage range. The low-voltage portion was measured with a finer sweep, while the higher-voltage portion used 0.01 V increments. The combined data contains 111 points. The low sweep extends through 0.93 V; the higher sweep begins at 0.95 V, so the 0.94 V point is not represented in the combined dataset.

Representative results are listed in Table 2.

Table 2: Representative frequency results for the nine-stage configuration.

V_{con} (V)	Frequency (MHz)	Frequency (GHz)
0.70	226.79	0.2268
0.80	356.82	0.3568
0.90	484.59	0.4846
0.93	521.40	0.5214
0.95	558.36	0.5584
1.00	618.47	0.6185
1.20	824.48	0.8245
1.40	974.43	0.9744
1.60	1075.50	1.0755
1.80	1135.71	1.1357

The nine-stage oscillator frequency increases from approximately 226.8 MHz at 0.7 V to 1.136 GHz at 1.8 V. As expected for a ring oscillator with more stages, this configuration operates at a lower frequency than the three-stage configuration for the same control voltage.

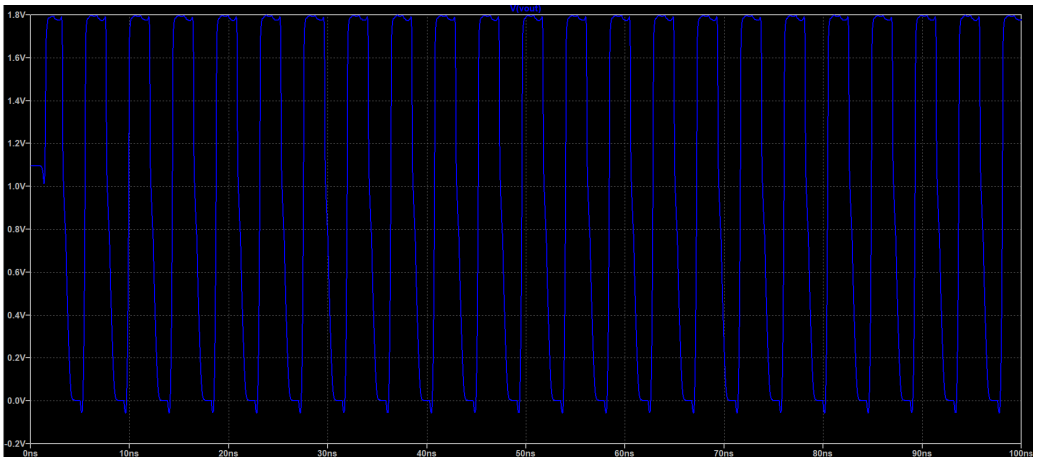


Figure 6: Transient output waveform of the nine-stage VCO at 0.7V control voltage.

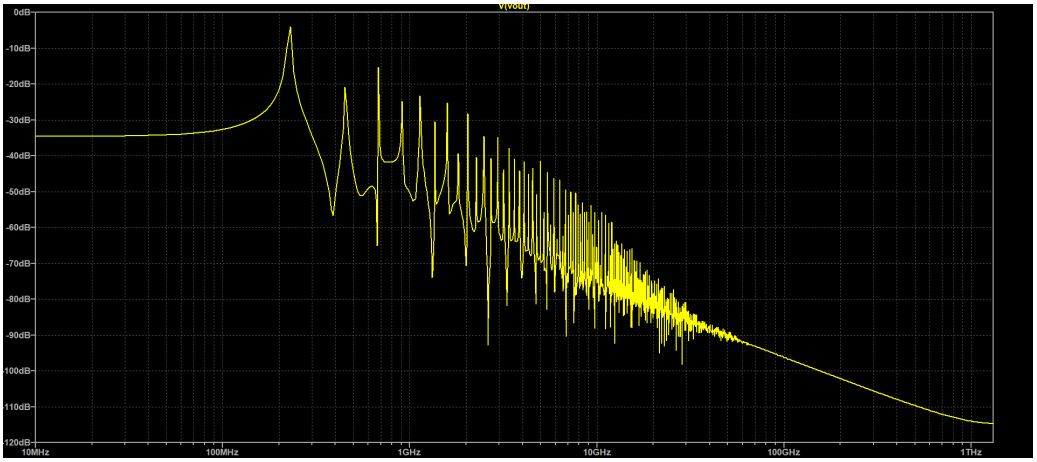


Figure 7: FFT analysis of the nine-stage VCO output.

5.4 Frequency versus control voltage

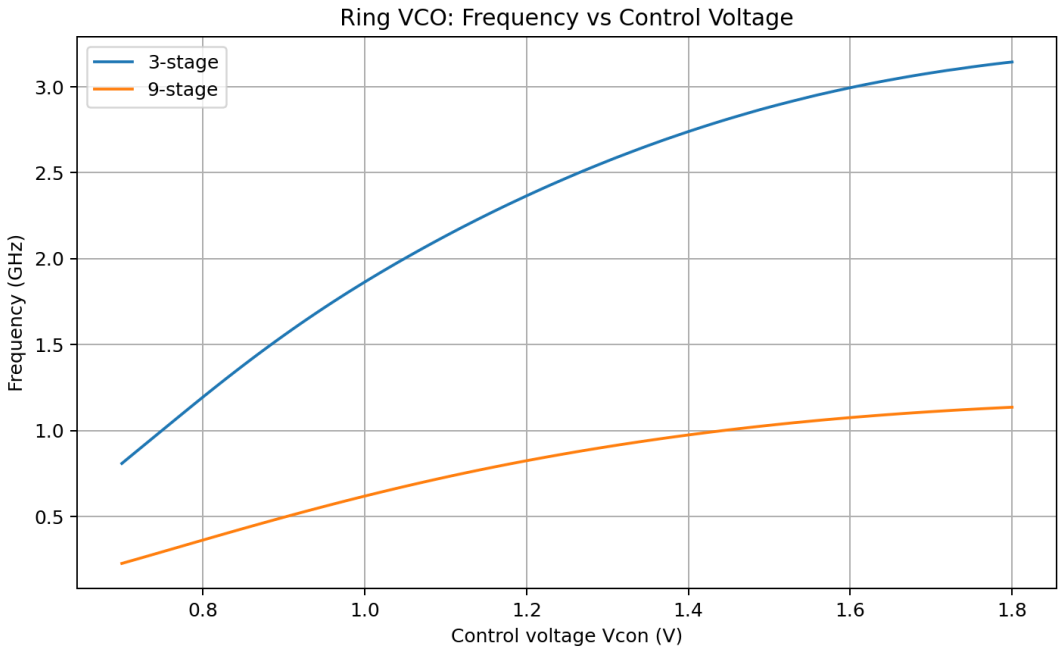


Figure 8: Frequency-versus-control-voltage plot comparing the three-stage and nine-stage configurations.

The measured frequency increases with V_{con} for both configurations. The three-stage configuration provides the higher-frequency range, whereas the nine-stage configuration extends operation toward lower frequencies. The relationship is nonlinear, which is consistent with the nonlinear current-voltage characteristics of MOS transistors and the dependence of inverter delay on bias conditions.

6 Power Consumption

6.1 Three-stage configuration

The measured average supply current increases from approximately $341.1\,\mu\mathrm{A}$ at $V_{\mathrm{con}} = 0.7\,\mathrm{V}$ to $1.181\,\mathrm{mA}$ at $1.8\,\mathrm{V}$. With a $1.8\,\mathrm{V}$ supply, the corresponding estimated average power rises from approximately $0.614\,\mathrm{mW}$ to $2.126\,\mathrm{mW}$.

6.2 Nine-stage configuration

The nine-stage power sweep shows average current increasing from approximately $141.55\,\mu\mathrm{A}$ to $553.67\,\mu\mathrm{A}$ over $V_{\mathrm{con}} = 0.7$–$1.8\,\mathrm{V}$. The corresponding estimated average supply power increases from approximately $0.2548\,\mathrm{mW}$ to $0.9966\,\mathrm{mW}$.

Table 3: Summary of measured frequency and supply-power ranges.

Configuration	f_{min} (MHz)	f_{max} (GHz)	P_{min} (mW)	P_{max} (mW)
Three-stage	808.46	3.1447	0.6140	2.1258
Nine-stage	226.79	1.1357	0.2548	0.9966

Note: The table reports endpoints from the measured sweeps. Power values for the nine-stage configuration are taken from its average-current sweep. The two configurations should be compared under the same simulation settings and loading assumptions.

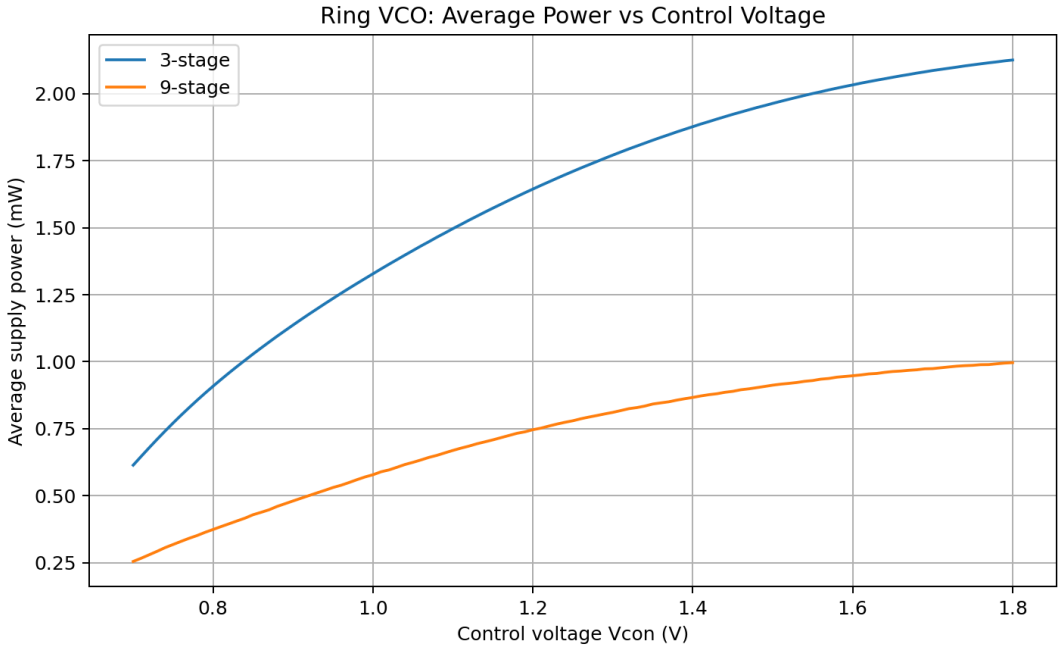


Figure 9: Average supply power versus control voltage for both configurations.

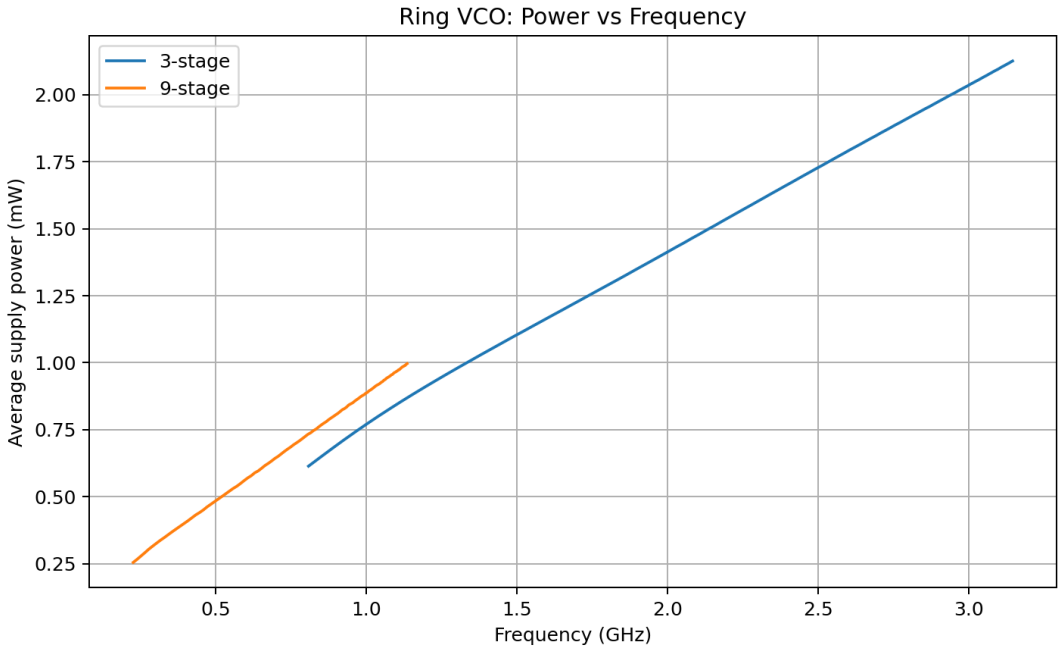


Figure 10: Average supply power versus oscillation frequency.

The power results show increasing supply power with increasing control voltage. This is expected because the available current and switching activity increase as the oscillator frequency rises. The three-stage configuration has a higher measured power in the current simulations, but the result depends on the implementation, number of stages, bias currents, and measurement

setup. “‘latex

7 Startup Delay Analysis

The startup delay of the ring VCO was measured from the beginning of the transient simulation until the output exhibited sustained oscillations. The measurements were performed for both the three-stage and nine-stage configurations at different control voltages, V_{con}.

7.1 Startup Delay Results

Table 4 summarizes the observed startup delays for the two configurations.

Table 4: Measured startup delay for the three-stage and nine-stage VCO configurations.

Configuration	V_{con} (V)	Startup Delay (ns)
Three-stage	0.7	3.7
Three-stage	1.8	1.6
Nine-stage	0.7	1.4
Nine-stage	1.8	1.0

For the three-stage configuration, the startup delay decreases from approximately 3.7 ns at $V_{\mathrm{con}} = 0.7\,\mathrm{V}$ to 1.6 ns at $V_{\mathrm{con}} = 1.8\,\mathrm{V}$. Similarly, for the nine-stage configuration, the startup delay decreases from approximately 1.4 ns to 1.0 ns over the same control-voltage range.

These observations indicate that the startup delay decreases with increasing control voltage for both configurations under the simulated conditions. The measured startup delay depends on the initial conditions, transistor biasing, and transient simulation setup.

7.2 Startup Waveforms

The transient waveforms used to determine the startup delay can be included in Fig. **??**. The startup interval should be visible in each plot, with the time at which sustained oscillations begin clearly indicated.

8 Control-Voltage Perturbation Experiment

An initial experiment was performed by adding a sinusoidal perturbation to the control voltage. An example behavioural-source expression was

```
V = 0.9 + 0.01*sin(2*pi*1Meg*time)
```

This represents a 1 MHz sinusoidal ripple with a 10 mV amplitude around a 0.9 V DC control level. Other perturbation amplitudes were also tried, including 1 mV, 5 mV, 10 mV, and 50 mV, at DC control levels of 0.7 V and 1.8 V.

The initial observation was that the recorded startup delay did not change substantially in several nine-stage trials, while some three-stage measurements varied. The recorded approximate startup delays are summarized below.

Table 5: Approximate startup-delay observations under a 1 MHz control-voltage perturbation. These are exploratory readings, not a jitter or phase-noise measurement.

Stages	DC V_{con} (V)	1 mV (ns)	5 mV (ns)	10 mV (ns)	50 mV (ns)
Nine	0.7	1.0	1.0	1.0	1.0
Nine	1.8	0.8	0.8	0.7	0.7
Three	0.7	0.7	0.7	1.6	1.4
Three	1.8	1.0	1.0	0.9	0.9

These startup-delay readings do not establish the oscillator's phase-noise or timing-jitter performance. The ripple period at 1 MHz is 1 μs, while a 100 ns transient captures only one-tenth of a ripple cycle. A more reliable study would simulate for several ripple periods, discard startup behaviour, and measure steady-state edge timing or cycle-to-cycle period variation. A controlled comparison with the perturbation disabled is also required.

9 Discussion

The simulations demonstrate that both selectable ring configurations oscillate and that the oscillation frequency increases with control voltage. The three-stage configuration reaches approximately 3.145 GHz, while the nine-stage configuration reaches approximately 1.136 GHz at the upper end of the tested range. The nine-stage design also reaches a lower measured frequency of approximately 226.8 MHz at 0.7 V.

The average supply power rises with the control voltage for both configurations. In the current simulations, the three-stage oscillator consumes more power than the nine-stage oscillator at the sweep endpoints. These figures should be treated as preliminary because they come from the present schematic and model setup, without extracted layout parasitics or a verified physical implementation.

The waveform inspections show periodic operation, but the measured rail excursions warrant further investigation. The added control-voltage ripple experiment was exploratory: the startup-delay observations do not establish a quantitative noise immunity, jitter, or phase-noise result. Similarly, the proposed output buffer is a future experiment and should not be described as a completed improvement until simulated.